

SOLUCIONARIO SEGUNDO PARCIAL

ASIGNATURA:

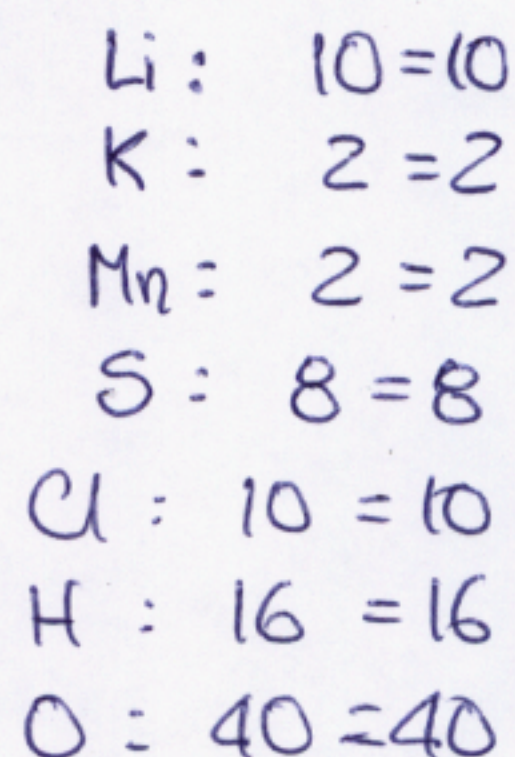
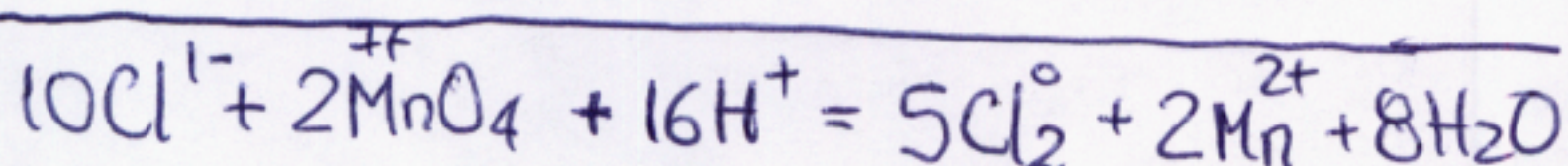
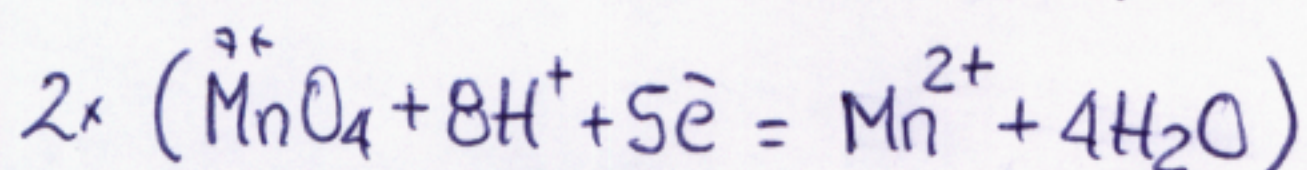
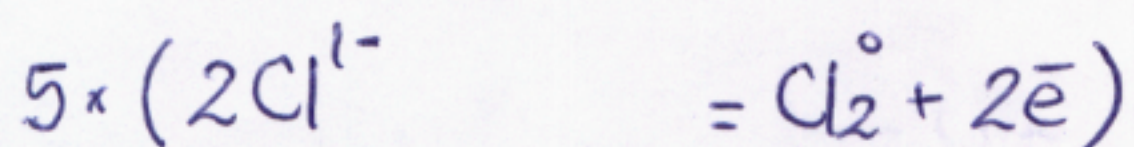
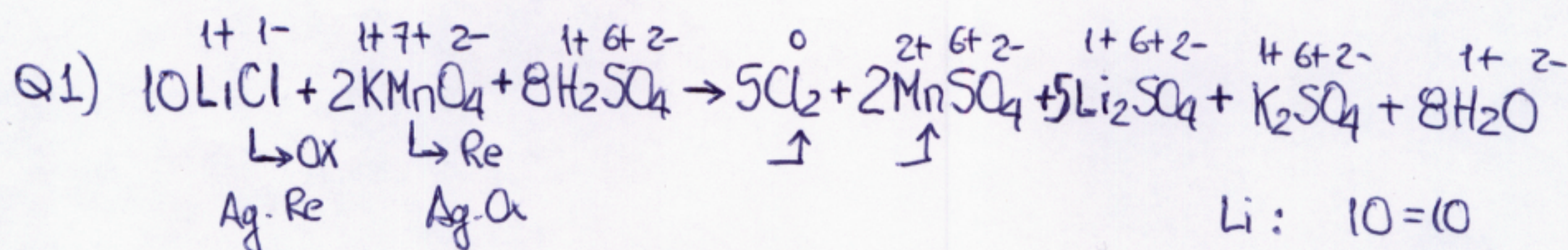
QUÍMICA

FECHA:

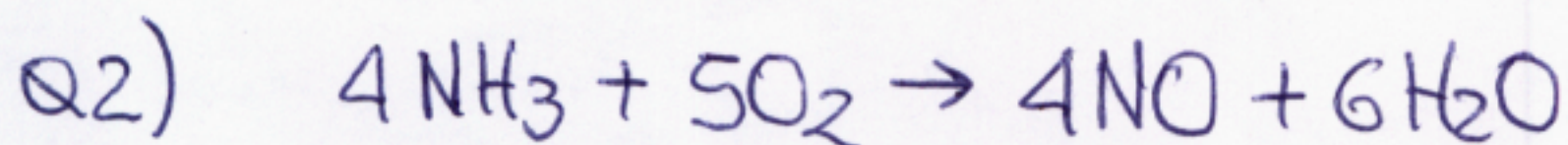
15/12/2025

PÁGINAS:

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$$Y = \frac{2 \times (2) - 10}{2^3 - 2(10)} = \frac{4 - 10}{8 - 20} = \frac{-6}{-12} \Rightarrow \boxed{Y = \frac{1}{2}} \text{ D)}$$

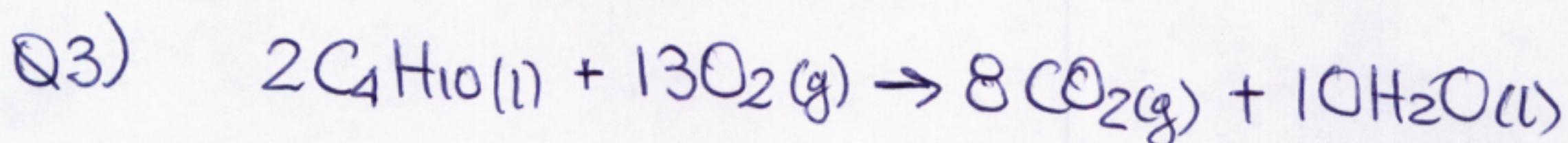


$$1048 \text{ L aire} \times \frac{21 \text{ L O}_2}{100 \text{ L aire}} = 220,08 \text{ L O}_2$$

$$200 \text{ L NH}_3 \times 4 = 50 \text{ L NH}_3 \quad \text{R.E}$$

$$220,08 \text{ L O}_2 \times 5 = 44,016 \text{ L O}_2 \quad \text{R.L}$$

$$220,08 \text{ L O}_2 \times \frac{4 \text{ L NO}}{5 \text{ L O}_2} = \boxed{176,1 \text{ L NO}} \text{ D)}$$



$$10 \text{ g C}_4\text{H}_{10} \times \frac{1 \text{ mol C}_4\text{H}_{10}}{58 \text{ g C}_4\text{H}_{10}} \times \frac{13 \text{ mol O}_2}{2 \text{ mol C}_4\text{H}_{10}} \times \frac{32 \text{ g O}_2}{1 \text{ mol O}_2} = \boxed{35,86 \text{ g O}_2} \text{ B)}$$

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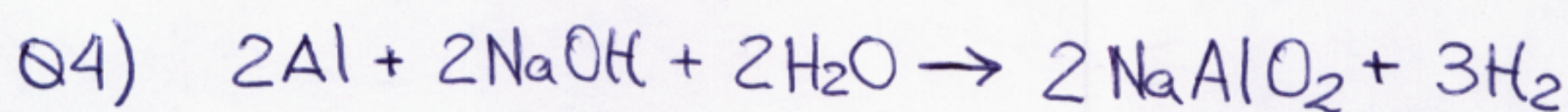
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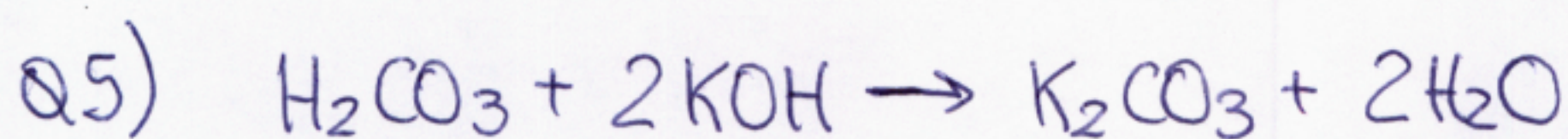
$$V_{\text{H}_2} = 1,8 \text{ L H}_2$$

$$P_{\text{H}_2} = 730 \text{ torr} - 40 \text{ torr} = 690 \text{ torr}$$

$$T_{\text{H}_2} = 34^\circ\text{C} + 273 = 307 \text{ K}$$

$$P_{\text{H}_2} \cdot V_{\text{H}_2} = n_{\text{H}_2} \cdot R \cdot T_{\text{H}_2} \Rightarrow n_{\text{H}_2} = \frac{P_{\text{H}_2} \cdot V_{\text{H}_2}}{R \cdot T_{\text{H}_2}} = 0,065 \text{ mol H}_2$$

$$0,065 \text{ mol H}_2 \times \frac{2 \text{ mol Al}}{3 \text{ mol H}_2} \times \frac{27 \text{ g Al}}{1 \text{ mol Al}} \times \frac{100 \text{ g muestra}}{64 \text{ g Al}} = \boxed{1,82 \text{ g muestra}} \quad \text{c)}$$



$$30 \text{ mL KOH} \times \frac{4 \text{ mol KOH}}{1000 \text{ mL KOH}} \times \frac{1 \text{ mol H}_2\text{CO}_3}{2 \text{ mol KOH}} = 0,06 \text{ mol H}_2\text{CO}_3$$

$$M = \frac{0,06 \text{ mol}}{0,05 \text{ L}} \Rightarrow \boxed{M = 1,2 \text{ M}} \quad \text{A)}$$